Biliary Atresia, Part Eleven: Nutrition and Growth. One Hundred and Thirty to One Hundred and Fifty Percent of Energy, Medium-Chain Fats, Branched-Chain Amino Acids, Tube Feeding and the Fat-Soluble Vitamins.

Imagine a four-month-old girl who had a Kasai operation for biliary atresia at forty-eight days of age. The operation drained some bile, but not enough. At three months her bilirubin is still five point eight milligrams per decilitre. Her weight, which sat a little below average at birth, has slid from a z-score of minus zero point eight to minus two point three, which means she has crossed from the lower part of the normal range to below the second centile. Her liver and spleen are large and there is a little fluid in her abdomen. Her clotting time, measured as the international normalised ratio, is one point six, and her vitamin D level is eleven nanograms per millilitre. Her mother is breastfeeding and giving a special formula as well, and she asks a simple question: why is my baby not growing when she feeds all day? The question for the clinician is how to feed this child so that she reaches liver transplantation in the best possible condition.

The main idea of this chapter is that, in an infant whose Kasai operation has not cleared the jaundice, nutrition is not a supportive extra. It is part of preparing the child for transplantation. Poor growth is a marker of a poor outcome, and it is also a risk factor that we can change. The chapter walks from the mechanism of malnutrition to the targets, and then to the tools: formulas rich in medium-chain fats, branched-chain amino acids, feeding by tube and by vein, and fat-soluble vitamins given one by one and adjusted to blood levels.

A few terms first. A z-score describes how far a measurement lies from the average for age and sex, in units of standard deviation. A z-score of zero is average, minus one is roughly the sixteenth centile, and minus two is roughly the second centile. Medium-chain triglycerides, usually shortened to M C T, are fats with fatty acid chains of eight to ten carbon atoms, unlike the long-chain fats that make up most of breast milk and food. The branched-chain amino acids are leucine, isoleucine and valine. The fat-soluble vitamins are vitamins A, D, E and K. And the Kasai operation, also called a portoenterostomy, joins a loop of bowel to the cut surface of the liver at the porta hepatis to let bile drain.

The evidence in this chapter comes mainly from the 2019 joint position paper on nutrition support of children with chronic liver disease from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, written by Mouzaki and colleagues. Growth and outcome data come from the Biliary Atresia Research Consortium, reported by DeRusso in 2007, and from the Studies of Pediatric Liver Transplantation registry, reported by Utterson in 2005. Vitamin data come from the Childhood Liver Disease Research Network, reported by Shneider in 2012 and Venkat in 2014. Feeding data come from a Lille cohort reported by

Privat in 2022, from a United Kingdom cohort of two hundred infants published in Clinical Nutrition in 2025, from two small studies of branched-chain amino acids published in 1992, and from a Denver series on parenteral nutrition reported by Sullivan in 2012. For comparison we will look at adult guidance from the European Association for the Study of the Liver in 2019 and the American Association for the Study of Liver Diseases in 2021, and at the 2026 paediatric transplant evaluation guideline from the American Association for the Study of Liver Diseases, the American Society of Transplantation and the North American paediatric society.

Part one. Five ways a cholestatic infant starves

Why does an infant with persistent jaundice fail to grow even when she seems to feed constantly? There are five separate reasons, and they act together.

The first is intake. These infants often feed poorly. A large liver and spleen, and sometimes fluid in the abdomen, press on the stomach, so the infant feels full early. Vomiting and fatigue at the breast are common. The second is absorption, and this is the central problem. Fat is digested in the small bowel with the help of bile acids, which form tiny droplets called micelles that carry fat and fat-soluble vitamins to the bowel wall. When bile does not reach the bowel, the concentration of bile acids falls below the level needed to form micelles. Long-chain fats and the vitamins A, D, E and K then pass through the gut without being absorbed. The infant loses a large part of her most energy-dense food in the stool.

The third reason is metabolism. A diseased liver stores little glycogen, the sugar reserve that carries a healthy infant through the hours between feeds. So after even a short fast, the infant switches early to burning fat and protein, and uses her own muscle as fuel. The fourth reason is expenditure. Inflammation, episodes of cholangitis and portal hypertension all raise the energy the body uses at rest. And the fifth is losses: fat in the stool, fluid into the abdomen, losses through diuretics and through bleeding, and loss of trace elements such as zinc.

There is one more trap. Weight overstates nutrition in these children. The enlarged liver, the enlarged spleen and the fluid in the abdomen add weight, while muscle and fat are quietly lost. That is why the arm is more honest than the scale. The mid-upper arm circumference, measured with a tape around the middle of the upper arm, and the triceps skinfold, measured with calipers at the back of the arm, reflect muscle and fat stores and are not changed by ascites. Length and head growth also matter. Each of the five causes has its own fix, so a feeding plan has to address all five rather than simply adding calories.

Part two. Growth predicts who will need a liver

How strongly does growth relate to outcome? The Biliary Atresia Research Consortium studied one hundred infants who had a Kasai operation at nine American centres between 1997 and 2000. They divided them into a good outcome group, fifty-four infants who were alive with their own liver at two years, and a poor outcome group, forty-six infants who had been

transplanted or had died by two years. Six months after the operation, the mean weight z-score was minus one point two in the good outcome group and minus two point one in the poor outcome group, a difference that was highly significant. Growth velocity for both weight and length was slower in the poor outcome group.

The more interesting finding came from infants whose bilirubin at three months was in the middle range, between two and six milligrams per decilitre, the group in which the outcome is hardest to predict. Even in these twenty-eight infants, weight at three months separated the groups: a z-score of minus one point zero in those who did well and minus two point zero in those who did not. In other words, a falling weight curve is a second marker, beside bilirubin, of who will need a liver.

Growth also follows the child into the operating theatre. The Studies of Pediatric Liver Transplantation registry analysed seven hundred and fifty-five children with biliary atresia listed for transplantation. Survival three years after transplantation was eighty-eight percent. Infancy and a paediatric end-stage liver disease score of twenty or more were the main risks for death on the waiting list. But after the transplant, a height or weight below minus two standard deviations was one of the risk factors for death and for loss of the graft. The authors concluded that poor nutrition should prompt aggressive support.

Muscle tells the same story. In a Canadian cohort of twenty-five children aged one to sixteen years waiting for a liver transplant, published in Liver Transplantation in 2021, forty percent had sarcopenia, defined as a total psoas muscle area on cross-sectional imaging more than two standard deviations below the mean. Children with sarcopenia spent a median of three and a half days in the paediatric intensive care unit after transplantation, compared with two days for those without it. The 2019 position paper cites an American study of thirty-five children with end-stage liver disease in whom muscle mass was twenty-three percent lower than in healthy controls, while visceral fat was sixty-nine percent higher. A child can look plump and still be losing muscle. And the 2026 paediatric transplant evaluation guideline recommends aggressive nutritional support before transplantation in children with cholestatic liver disease to improve outcomes. The practical lesson is that the waiting time is the window to rebuild muscle, and that in older children the psoas area on the pre-transplant scan is a free measurement worth asking the radiologist for.

Part three. What to measure, and what to aim for

The position paper gives a clear monitoring frame. In an infant with cholestasis, weight, length and head circumference, together with mid-upper arm circumference and triceps skinfold, should be measured every two to four weeks and expressed as z-scores. Dietary intake should be recorded at each visit and expressed as a percentage of the energy target, because the gap in intake appears before growth falls. Albumin and other protein markers are checked about every three months, remembering that a low albumin reflects the liver's ability to make protein as much as nutrition. Vitamins A, D, E and K are measured every three to six months, and

monthly in severely cholestatic infants. Essential fatty acids are checked every three to six months, and zinc and iron as needed. A useful rule from the Childhood Liver Disease Research Network is that a total bilirubin above two milligrams per decilitre predicts vitamin deficiency better than serum bile acids do.

Now the targets. The position paper recommends energy of at least one hundred and thirty percent of the normal requirement for age, and many centres aim for one hundred and thirty to one hundred and fifty percent. Resting energy expenditure can be measured by indirect calorimetry where available, but in practice the target is adjusted to growth rather than fixed to a formula. Protein should be one hundred and thirty to one hundred and fifty percent of the age requirement and should never be restricted. Encephalopathy is rare in infants, and restricting protein only speeds the loss of muscle. Fat should provide thirty to fifty percent of energy, because fat is the densest fuel. Of that fat, the medium-chain share should start at about thirty percent, can rise to about fifty percent in tube feeds, and should stay below eighty percent. Linoleic acid should provide at least three percent of energy and alpha-linolenic acid between zero point seven and one percent, so that essential fatty acids are not lost. Carbohydrate provides forty to sixty percent of energy, and glucose polymers can add energy without adding volume. Finally, fasting must be short, and continuous overnight feeds are used if the infant becomes hypoglycaemic. To give a sense of scale, in the Lille cohort of infants fed by tube, total intake at the start of tube feeding was one hundred and eighty-six percent of reference values.

Part four. Medium-chain fats: a shortcut around the missing bile

Why are medium-chain fats used at all? Fats with chains of eight to ten carbons can be broken down by pancreatic enzymes without bile, and they are absorbed directly into the portal vein rather than into the lymph. So they bypass the step that fails in cholestasis. They provide about eight point three kilocalories per gram, a little less than the nine kilocalories of long-chain fat.

But medium-chain fats have limits. They contain no essential fatty acids, so a diet too rich in them can cause essential fatty acid deficiency, which shows up as a rise in the ratio of triene to tetraene fatty acids in the blood, as well as dry skin and poor growth. They do not carry the fat-soluble vitamins. And at high doses they cause diarrhoea.

The evidence is thinner than the habit. The first report, by Cohen and Gartner in 1971, described eleven children with biliary atresia given diets rich in medium-chain fat. Faecal fat fell, stools improved, three children caught up in growth, and three who started in early infancy grew normally. That is the foundation on which decades of practice rest. The largest modern study, a United Kingdom cohort of two hundred infants after the Kasai operation published in Clinical Nutrition in 2025, asked whether more medium-chain fat meant better growth. The prescribed formula was on average fifty-seven percent medium-chain fat, and the median intake was two point seven grams per kilogram per day. Infants with the higher intake took more energy, one hundred and eighteen versus one hundred and eight kilocalories per kilogram

per day. Yet at three months their weight z-score was lower, minus one point seven versus minus one point zero, and their length z-score was lower too, minus one point three versus minus zero point six. For each extra gram per kilogram per day, the weight z-score fell by about a quarter. After three months there was no difference in growth, and there was no link between medium-chain fat and nutritional status or clinical outcome at two years. Because this was an observational study, it cannot prove that medium-chain fat slows growth. The likely explanation is that sicker, slower-growing infants were prescribed more of it. The authors call for a randomised trial.

The honest summary is that medium-chain fat makes physiological sense and is used everywhere, but no randomised trial shows that a higher share improves growth or survival in biliary atresia. Total energy delivered matters more than the blend of fat. A sensible approach is to continue breastfeeding, add a formula rich in medium-chain fat, or use a formula with fifty to sixty percent medium-chain fat if breast milk is not available, and when growth falters, ask first whether the problem is the amount of energy rather than the type of fat.

Part five. Branched-chain amino acids

Leucine, isoleucine and valine are unusual among amino acids because they are broken down mainly in muscle rather than in the liver. Their levels in blood fall in end-stage liver disease, and supplementing them may spare muscle protein. That is the rationale.

The paediatric evidence consists of two small studies from 1992. In Birmingham, Charlton and colleagues fed ten malnourished children with advanced cirrhosis for eight weeks by nasogastric tube, using a whey protein enriched with branched-chain amino acids, a fat blend of thirty-four percent medium-chain and sixty-six percent long-chain fat, and glucose polymer. Six of them first had an eight-week control period in which they stayed malnourished despite high intake. During tube feeding every anthropometric measure improved, no child developed encephalopathy, and ammonia rose only slightly. In Brisbane, Chin and colleagues ran a randomised crossover trial comparing a formula enriched with branched-chain amino acids with a standard formula at the same energy, and found better gains in body composition with the enriched formula.

Adult guidelines disagree on this point. The 2019 European guideline on nutrition in chronic liver disease supports long-term branched-chain amino acid supplements in advanced cirrhosis. The 2021 American practice guidance on malnutrition, frailty and sarcopenia does not recommend them and asks instead that the daily protein target be met from varied sources. The 2019 paediatric position paper mentions a formula enriched with branched-chain amino acids but gives no dose. A reasonable position for children is this: in a sarcopenic infant awaiting transplantation who is already at the energy and protein targets, a semi-elemental formula enriched with branched-chain amino acids is defensible, but it is never a substitute for enough total protein.

Part six. Feeding by tube, and by vein

When should a feeding tube be placed? The position paper says that if energy goals cannot be met for more than two weeks, or if two to four weeks of falling weight, arm or length measurements occur, enteral feeding by tube should begin. The tube should be a fine-bore nasogastric tube, or nasojejunal if the infant vomits. Oesophageal varices are not a contraindication to a soft, fine tube. The usual pattern keeps daytime oral feeds, so that the infant keeps practising sucking and swallowing, and adds continuous overnight feeds by pump, which deliver energy and prevent hypoglycaemia. The formula is usually semi-elemental, rich in medium-chain fat, and concentrated to between one and one and a half kilocalories per millilitre. One thing to avoid is a percutaneous gastrostomy in a child with portal hypertension, because varices can form around the stoma and bleed, and ascites raises the risk of peritonitis.

Does tube feeding work? The Lille team in France reviewed thirty-one children with biliary atresia fed by tube while waiting for a transplant between 1995 and 2018. Tube feeding began at a median age of seven months and lasted a median of nine months, mostly with a semi-elemental formula containing sixty percent medium-chain fat. Over that time the median weight-for-age z-score improved from minus one point six to minus zero point five, length-for-age from minus one point three to minus zero point four, and body mass index from minus one point two to minus zero point two. By the end, ninety-four percent had a weight-for-length z-score above minus two. Twenty-three percent had complications, mostly nausea, vomiting, diarrhoea or the tube coming out, and ten percent stopped because of a major complication. But here is the sobering point: fifty-five percent were eventually switched to parenteral nutrition, mainly because weight gain was insufficient or because of ascites. So tube feeding is effective and safe when started early, but for more than half of these children the gut alone was eventually not enough.

Parenteral nutrition, meaning nutrition given into a vein, is indicated when enteral feeds are not tolerated or do not meet growth targets, or when ascites limits the volume that can be given. In Denver, Sullivan and colleagues reviewed twenty-five listed children with biliary atresia who received supplemental parenteral nutrition and twenty-two who did not, and the children on parenteral nutrition gained arm circumference and skinfold thickness. The risks are real: a central line in an infant already prone to cholangitis brings infection risk, and parenteral nutrition can add its own cholestasis. The safest pattern keeps some enteral feeding going, uses a mixed lipid emulsion with cycled infusions, and starts in hospital. Parenteral nutrition is a bridge to a graft, not a home therapy for years, and the decision should be shared with the transplant team.

Long tube feeding has one more cost. Infants who are fed by tube for months lose oral motor practice. Small tastes by mouth, non-nutritive sucking and early involvement of a feeding therapist all help, and parents should be told that feeding aversion after transplantation is common and treatable.

Part seven. The fat-soluble vitamins

The Childhood Liver Disease Research Network studied ninety-two infants after the Kasai operation, all of whom received a liquid multivitamin preparation made with tocopheryl polyethylene glycol succinate, a water-soluble form of vitamin E usually called T P G S, together with extra vitamin K. Levels were measured one, three and six months after surgery. Despite supplementation, biochemical insufficiency was common: vitamin A in twenty-nine to thirty-six percent of infants across the time points, vitamin D in twenty-one to thirty-seven percent, vitamin K in ten to twenty-two percent and vitamin E in sixteen to eighteen percent. Insufficiency was far more common when the bilirubin was two milligrams per decilitre or more. Among infants with a bilirubin above two, only three of twenty-four were sufficient in all four vitamins at three months, and none of twenty-three at six months. The message is that a combined multivitamin is a starting dose, not a plan. Once bilirubin is above two, each vitamin should be given separately and adjusted to its own level.

The position paper gives starting doses. For vitamin A, five thousand international units a day under ten kilograms and ten thousand above ten kilograms, adjusted to levels; the best marker of deficiency is a molar ratio of retinol to retinol-binding protein below zero point eight, and too much vitamin A can cause raised intracranial pressure and liver injury, so it must be monitored. For vitamin D, cholecalciferol two thousand to five thousand international units a day, with some centres using fifty thousand units weekly; a twenty-five-hydroxy vitamin D level below twenty nanograms per millilitre signals insufficiency, and rickets and fractures are the consequence. For vitamin E, the water-soluble form at fifteen to twenty-five international units per kilogram per day; the serum level alone can look falsely normal when blood lipids are high, so it is judged as a ratio of vitamin E to total lipids, with a value below zero point six milligrams per gram in infants indicating deficiency. A useful trick is that this form of vitamin E forms its own micelles, and giving it together with vitamins A and D improves their absorption too. For vitamin K, two to five milligrams a day by mouth, with one to ten milligrams intravenously if the clotting time stays high; reactions to intravenous vitamin K have been reported. The clotting time also reflects the liver's ability to make clotting factors, so a high value that corrects after intravenous vitamin K points to deficiency, while one that does not correct points to liver failure.

Vitamin K deserves a special word because bleeding into the brain can be the first sign of biliary atresia. A 2025 survey from the Japanese Biliary Atresia Registry compared two ways of giving vitamin K prophylaxis in the first months of life. With weekly oral vitamin K until three months of age, no infant had vitamin K deficiency bleeding, while bleeding was more frequent with the three-dose schedule. The authors concluded that the three-month method is effective and that early diagnosis is crucial.

Part eight. Sugar, salt, bone and trace elements

A few smaller items change growth and safety. Glucose: keep fasting intervals short, use continuous overnight feeds, and check glucose whenever feeds stop, for instance before a procedure. Sodium: about one to two millimoles per kilogram per day, restricted when the infant is fluid overloaded or has ascites, while watching for losses caused by diuretics. Bone: rickets and fractures follow low vitamin D, so calcium and phosphate intake must meet the norms for age. Zinc and iron are lost in stool and through bleeding, and zinc deficiency blunts appetite and growth. The water-soluble vitamins are usually adequate with a standard infant multivitamin. And every admission for cholangitis is a nutritional event: feeding stops while needs rise, so nasogastric feeds should restart as soon as the infant is stable rather than waiting for oral intake to recover.

Part nine. What the guidelines say

How do the documents compare? On the energy target, they agree in spirit: the paediatric position paper asks for at least one hundred and thirty percent of the age requirement, adult guidance asks for thirty to forty kilocalories per kilogram per day, and the 2026 paediatric transplant guideline asks for aggressive support before transplantation. On protein, they agree that it should not be restricted. On fasting, they agree: the paediatric paper asks for frequent feeds and overnight tube feeding, and adult guidance recommends a late-evening snack and no more than three to four hours without food. On branched-chain amino acids, they differ, with the European adult guideline in favour, the American adult guidance against, and the paediatric paper silent on dose. On how to measure sarcopenia, there is a gap: adults have computed tomography muscle indices and frailty tools, while children have only the psoas area without agreed cut-offs. And on the medium-chain fat share and monitoring of essential fatty acids, the difference is material, because this is a paediatric question that adult guidance does not address and that no trial has answered.

The material gap across all of this is simple to state. No randomised trial supports the medium-chain fat share, the branched-chain amino acid formula or the one hundred and thirty to one hundred and fifty percent energy target in biliary atresia. All three rest on physiology and small series.

Part ten. Traps, pearls and decisions

There are several traps. The first is a weight that looks fine because organomegaly and ascites hide wasting; measure the arm. The second is reassurance because a multivitamin has been prescribed; combined drops do not correct deficiency once bilirubin is above two, so check levels. The third is raising the medium-chain fat share instead of raising energy, with essential fatty acid deficiency as the reward above eighty percent of fat. The fourth is placing a gastrostomy in a child with portal hypertension. The fifth is restricting protein out of fear of ammonia, which accelerates sarcopenia. And the biggest trap of all is waiting. A feeding tube placed after three months of decline buys back less muscle than one placed after three weeks, and a child who reaches transplantation with a weight or length below minus two standard

deviations carries that risk into the post-operative course.

The numbers worth remembering are these. Six months after the Kasai operation, the weight z-score was minus two point one in infants who were transplanted or died and minus one point two in those who survived with their own liver. A height or weight below minus two standard deviations predicted death and graft loss after transplantation among seven hundred and fifty-five children. The energy target is at least one hundred and thirty percent of the age requirement, with protein at one hundred and thirty to one hundred and fifty percent and never restricted. Medium-chain fat starts at about thirty percent of fat, rises to fifty or sixty percent in tube feeds, and stays below eighty percent. Vitamin insufficiency after the Kasai operation affected up to about a third of infants despite supplements, and with a bilirubin above two, none of twenty-three infants was sufficient in all four vitamins at six months. Tube feeding improved weight-for-age from minus one point six to minus zero point five over nine months. And sarcopenia was found in forty percent of children on the waiting list.

Let us return to five common situations. For an infant six weeks after the Kasai operation whose bilirubin is falling and who is gaining well, continue breast milk with a formula rich in medium-chain fat, give vitamins A, D, E and K separately, and remeasure in two to four weeks, because prevention is cheaper than repletion. For an infant of three months with a bilirubin of five and a weight z-score of minus two point three, raise energy to one hundred and thirty to one hundred and fifty percent, start overnight nasogastric feeds, and open the transplant discussion, because growth failure predicts a poor outcome. For an infant with a clotting ratio of one point six before a liver biopsy, give intravenous vitamin K and recheck, then continue daily oral vitamin K, so that deficiency is separated from synthetic failure. For an infant with ascites and varices whose family asks about a gastrostomy, keep the nasogastric tube and avoid the gastrostomy. And for a listed child whose arm measurements keep falling despite full tube feeding, start supplemental parenteral nutrition in hospital together with the transplant team, as a bridge to the graft.

For the four-month-old girl at the start of this chapter, the plan follows directly. Her energy target rises to about one hundred and forty percent, with breast milk kept and a concentrated formula rich in medium-chain fat given by nasogastric pump overnight. Her vitamins are separated and dosed to levels, starting with vitamin D and intravenous then oral vitamin K. Her arm circumference and skinfold are recorded every two weeks. And because her bilirubin and her growth both point the same way, her transplant assessment begins now rather than after another month of decline.

What should you do differently in clinic? Write weight, length, arm circumference and triceps skinfold as z-scores in every letter, together with the current energy intake as a percentage of target and the last vitamin levels with their dates. Treat a falling arm measurement as the earliest warning you can act on. Once bilirubin is above two milligrams per decilitre after the Kasai operation, stop relying on a combined multivitamin and dose each vitamin to a level. Think about energy before fat type. Place a fine-bore nasogastric tube early, within two to four

weeks of faltering growth, and never a gastrostomy when there is portal hypertension. Do not restrict protein. And remember that in biliary atresia, the months on the waiting list are not empty time: they are the window in which nutrition can make the operation safer.